

Resilient India AI Autonomous Geopolitical Risk Intelligence & Procurement Orchestration Engine (Problem Statement 2)

Turning Reactive Energy Crisis Response into Anticipatory National Defense

TEAM RESILIENT INDIA

ENERGY SUPPLY CHAIN RESILIENCE

HACKATHON 2026





India's Energy Grid is Built on Fragile Maritime Chokepoints



Import Dependency

India imports almost all crude oil to sustain domestic demand.

Strait of Hormuz

National supply transits through this single, volatile geopolitical chokepoint.



SPR Buffer

Strategic Petroleum Reserves offer minimal cushion during naval blockades.

Planning Lag

Traditional systems take 47 days longer to stabilize after energy shocks.

 **Core Vulnerability:** Traditional planning tools cannot model geopolitical threats in real time.

ResilientIndia AI: Real-Time Risk Intelligence & Orchestration

An AI-powered intelligence platform that turns geopolitical signal detection into executable procurement recommendations in **under 3 seconds**.



Geopolitical Risk Intelligence

Continuous RAG-backed signal detection ingesting news feeds, AIS vessel tracking, and sanctions registries.



Multi-Agent Orchestration

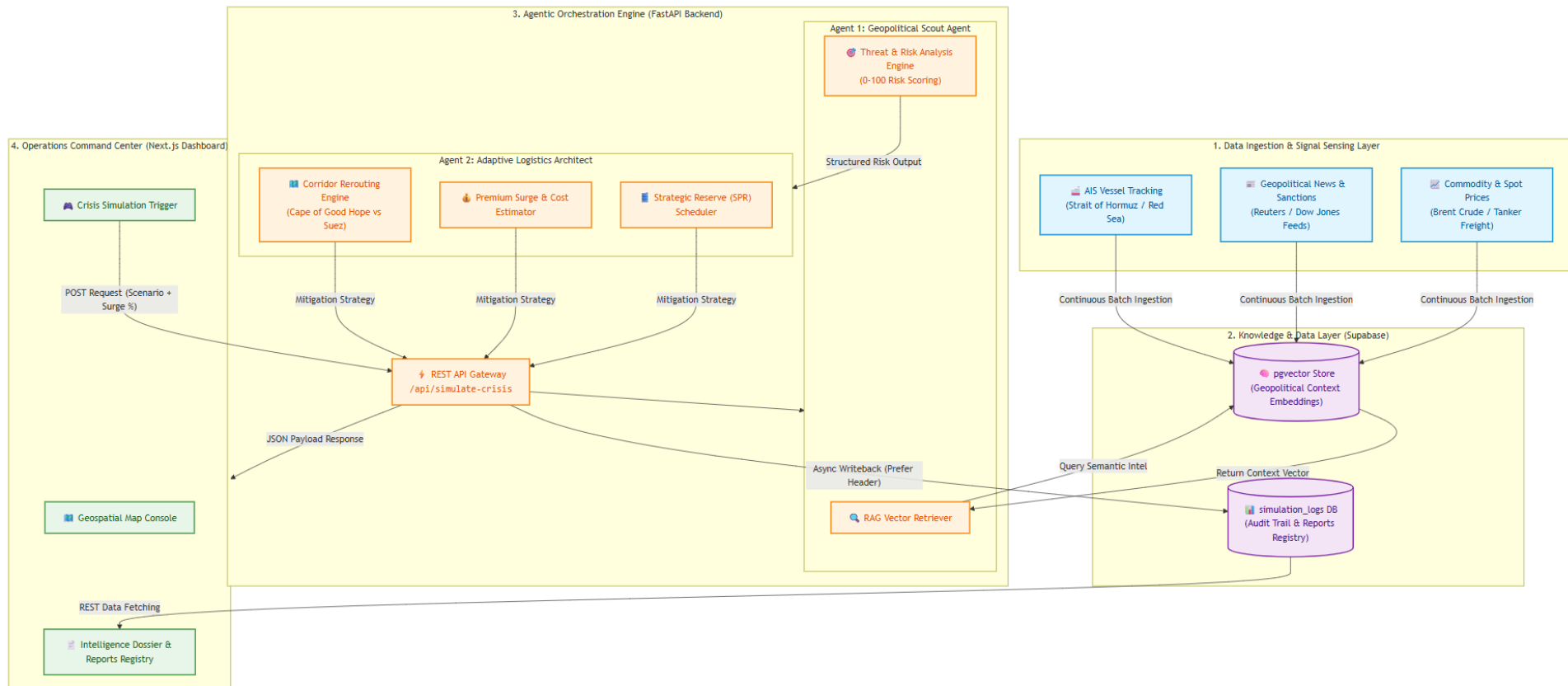
Specialized AI agents calculate threat risk scores (0–100) and design actionable mitigation plans.



Strategic Reserve Optimization

Automated drawdown models for national underground crude caverns at Visakhapatnam, Mangaluru, and Padur.

Decoupled Engine Flow & Production Architecture



Dual-Agent Intelligence Pipeline in Action



Agent 1: Geopolitical Scout (Assessment)

Performs vector similarity search (RAG) over geopolitical context stores. Translates unstructured news and blockade feeds into an objective **Risk Score (0–100)** and primary threat classification.

Agent 2: Adaptive Logistics Architect (Mitigation)

Corridor Rerouting: Compares travel delays and safety factors (Cape of Good Hope vs. Suez Canal vs. Holding in Berth). **Premium Surge Estimator:** Calculates freight surcharges, war-risk insurance spikes, and spot market premiums. **SPR Scheduler:** Computes precise crude drawdown rates to bridge supply gaps without exhausting reserves prematurely.



Real-Time Intelligence Dossiers & Audit Ledger



Interactive Stress Testing

Operators input custom crisis scenarios and premium surge scale percentages to test supply chain resilience.



Instant Ledger Persistence

Every simulation run automatically writes back to Supabase with UTC timestamps and explicit API headers.



Security-Classified Dossiers

Generates executive-ready Markdown dossiers tagged with official security tags: **SECRET, CONFIDENTIAL, RESTRICTED.**

Measuring the Value of Anticipatory Intelligence

Capability	Traditional	ResilientIndia AI
Response Latency	47 Days Avg Lag	< 3 Seconds
Risk Signal Detection	Manual & Weekly	Continuous RAG Sensing
Procurement Alternatives	Reactive Spot Panic	Dynamic Rerouting
Compliance & Audit	Disparate Spreadsheets	Immutable Ledger & Dossiers

✔ **Primary Outcome:** Prevents refinery run-rate collapses, safeguards domestic fuel prices, and saves refiners millions in spot market panic premiums.

Expanding Across Strategic National Supply Networks



LNG & LPG Networks

Extending corridor risk scoring to natural gas pipelines and cryogenic ocean routes.



Critical Minerals & Coal

Mapping maritime supply routes for thermal coal and rare-earth metals.



Public Sector Refiners

Direct API integration with Indian Oil (IOCL), Bharat Petroleum (BPCL), and Hindustan Petroleum (HPCL).



Policymaker Portals

Direct decision-support feeds for national defense and energy ministry officials.

